

Mass, Energy, and Nuclear Reactions

NE 630

Lesson 01

Prepare: Read FNRP Sections 1.1–1.3. Define the key terms in your notes and use the learning objectives to guide the reading.

Macro objective

Determine and quantify the reactants, products, and energy inputs/outputs of nuclear reactions.

Vocabulary: nuclide; atomic number; atomic mass; mass number; nucleon; rest mass; total energy; kinetic energy; binary reaction; radioactive decay; Q value; binding energy; energy density

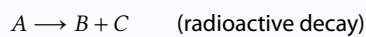
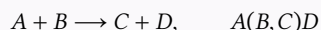
1 Nuclides and reaction language

Read the symbol

$${}^A_ZX, \quad A = Z + N$$

X
 Z
 A
 N

Reaction forms used in this lesson



Organize the initial and final values before using mass data:

	initial	final
mass number		
atomic number		

2 Mass change and Q values

Relations and conversion

$$\Delta E = \Delta mc^2, \quad Q = \left(\sum m_{\text{initial}} - \sum m_{\text{final}} \right) c^2$$
$$1 \text{ u} = 931.5 \text{ MeV}/c^2$$

The reaction rearranges bound nucleons (or electrons in a chemical reaction). Nuclear changes are typically on the MeV scale; chemical changes are typically on the eV scale. **Board check:** If $Q > 0$, which mass sum is larger, and what does that imply for a decay?

3 Relating neutron energies

Exact and classical expressions

$$E = mc^2, \quad m = \frac{m_0}{\sqrt{1 - (v/c)^2}}, \quad \beta = \frac{v}{c}$$

$$K_{\text{true}} = m_0 c^2 \left(\frac{1}{\sqrt{1 - \beta^2}} - 1 \right), \quad K_{\text{classical}} = \frac{1}{2} m_0 v^2$$

$$\% \text{ err} = 100 \frac{K_{\text{classical}} - K_{\text{true}}}{K_{\text{true}}}$$

Neutron data: $m_0 = 1.008664 \text{ u}$, $m_0 c^2 \approx 939.57 \text{ MeV}$, $c = 2.9979 \times 10^8 \text{ m/s}$.

Speed associated with a given K_{true}

Starting from the exact kinetic energy, isolate β^2 :

$$\beta^2 = 1 - \left(\frac{m_0 c^2}{K_{\text{true}} + m_0 c^2} \right)^2 = \frac{v^2}{c^2}.$$

Use this result to express $K_{\text{classical}}$ entirely in terms of K_{true} and $m_0 c^2$.

Threshold question

Find the maximum true neutron kinetic energy for which the magnitude of the classical error is no more than 1%.

Board result: $K_{\text{true}} \approx \dots\dots\dots$

Homework bridge: a 2.0 MeV fission neutron

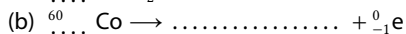
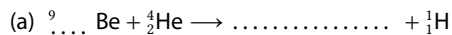
β^2		v	
$K_{\text{classical}}$		error	

Interpret the approximation

The classical expression the exact kinetic energy. Connect the 1% threshold to the lesson objective that neutron kinematics below about 10 MeV can often be treated classically.

4 Reaction completion and Q -value practiceComplete each reaction; then compute Q

Use one reaction as the board example; retain the others for practice.



Q_a Q_b

Q_c Q_d

Atomic-mass data used in the assigned problems (u)

nuclide	mass	nuclide	mass
${}^1\text{H}$	1.007825031898	${}^4\text{He}$	4.00260325413
${}^7\text{Li}$	7.01600343426	${}^9\text{Be}$	9.012183062
${}^{10}\text{B}$	10.012936862	${}^{12}\text{B}$	12.014352638
${}^{13}\text{C}$	13.00335483534	${}^{60}\text{Co}$	59.933815536
${}^{60}\text{Ni}$	59.930785129	n	1.008664

Atomic masses and decay

Before evaluating the β^- -decay Q value with the supplied *atomic* masses, show which electron masses are already contained in the parent and daughter entries.

Mass defect and binding energy per nucleon

Reconstruct the definitions from Sections 1.1–1.3:

$\Delta m =$

$B =$

$B/A =$

Use ${}^4_2\text{He}$ and the fixed mass data as a quick check.

5 Energy density of reactions

Plant-scale fixed inputs from the board notes

	nuclear plant	coal plant
electric power	1 GW _e	1 GW _e
efficiency	0.30	0.40
material basis	100 t UO ₂ over 5 y	10,000 t per day

Energy per unit material mass

Use $E = P \Delta t$ and account for thermal efficiency:

$$\left. \frac{E}{m} \right|_{\text{nuc}} \approx \frac{1 \times 10^9 \text{ W}_e}{0.30} \frac{5(365)(24) \text{ h}}{100 \times 10^3 \text{ kg}} = \dots$$

$$\left. \frac{E}{m} \right|_{\text{coal}} \approx \frac{1 \times 10^9 \text{ W}_e}{0.40} \frac{24 \text{ h}}{10,000 \times 10^3 \text{ kg}} = \dots$$

Ratio and interpretation:

Reaction-scale mass conversion

$$\left. \frac{\Delta m}{m_0} \right|_{\text{fission}} \approx \frac{200}{236(931.5)} = \dots$$

$$\left. \frac{\Delta m}{m_0} \right|_{\text{combustion}} \approx \frac{4 \times 10^{-6}}{44(931.5)} = \dots$$

How do these fractions explain the plant-scale comparison?

Three related scales

For pure matter,

$$\frac{E}{m} = c^2 \approx \dots$$

From the board notes,

$$\left. \frac{P}{m} \right|_{\text{nuc}} \approx \dots, \quad \left. \frac{P}{m} \right|_{\text{coal}} \approx \dots$$

Why is the power-density contrast smaller than the energy-density contrast?